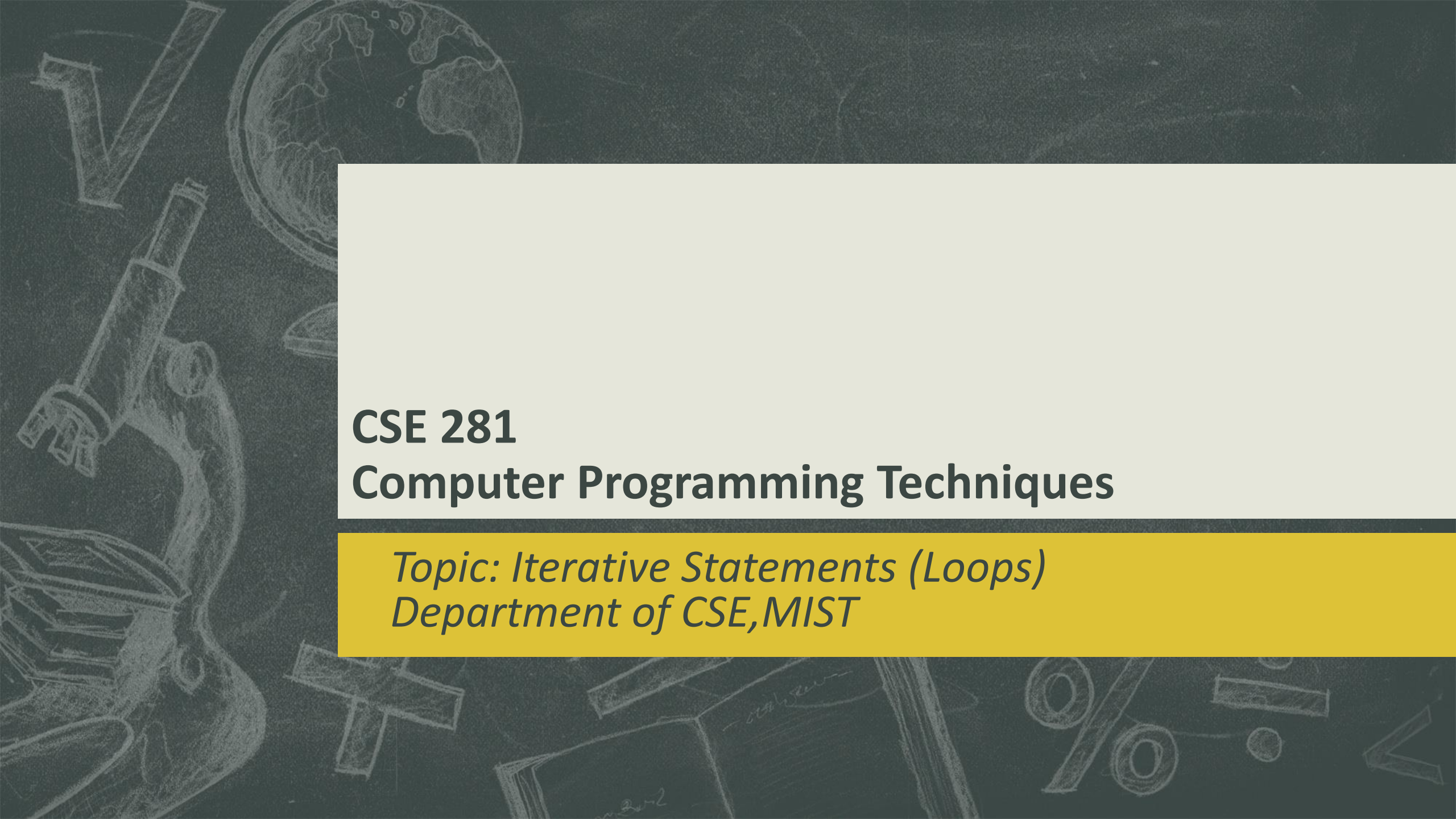


CSE 281
Computer Programming Techniques

Topic: Iterative Statements (Loops)
Department of CSE,MIST

Why are Loops used?

- **Write a code to print hello world**

```
#include <stdio.h>
int main() {
    printf( "Hello World\n");
        }
```

Why Loop is used?

- **Write a code to print hello world 10 times**

```
#include <stdio.h>
int main() {
    printf( "Hello World\n");
    printf( "Hello World\n");
    printf( "Hello World\n");
    printf( "Hello World\n");
    printf( "Hello World\n");
    printf( "Hello World\n");
    printf( "Hello World\n");
    printf( "Hello World\n");
    printf( "Hello World\n");
    printf( "Hello World\n");
    printf( "Hello World\n");  }
```

Why Loop is used?

- **Loops** in C programming is **used** when we need to **execute a block of statements repeatedly**.
- For example: Suppose we want to print “Hello World” 10 times. How can this be done efficiently?
- Using **loop** statement, we could easily solve this problem **without writing** the printf statement 10 times.

Solution 1:

```
#include <stdio.h>
int main() {
    printf( "Hello World\n");
    printf( "Hello World\n");
    printf( "Hello World\n");
    printf( "Hello World\n");
    printf( "Hello World\n");
    printf( "Hello World\n");
    printf( "Hello World\n");
    printf( "Hello World\n");
    printf( "Hello World\n");
    printf( "Hello World\n");
    printf( "Hello World\n");
     }
```

Why Loop is used?

- In looping, a sequence of statements are executed until some conditions for termination of the loop are satisfied.
- In short, in loop statement, we can **specify** how many times a block of statement will be executed.
- There are three loop statements in C:
  - for loop
  - while loop
  - do…while loop

for Loop - syntax

- **For loop** is used to repeat a statement or a block of statements a specific number of times.
- **Syntax:**

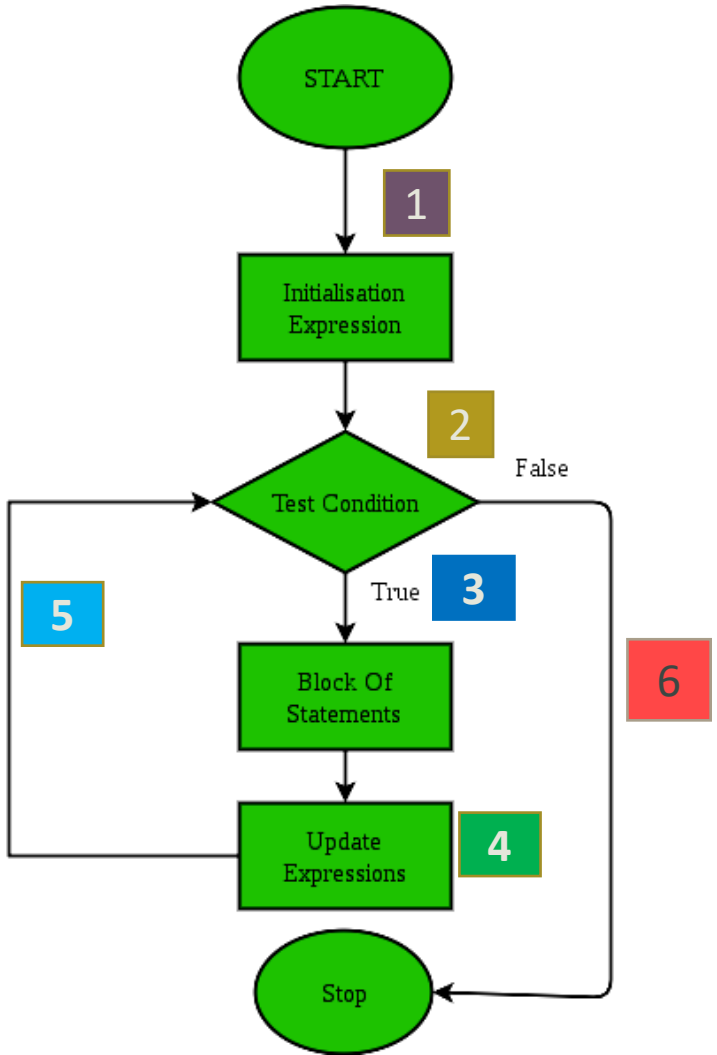
```
for (initialization of loop variable; conditional test; update loop variable) {
        // body of the loop
        // statements we want to execute
}
```

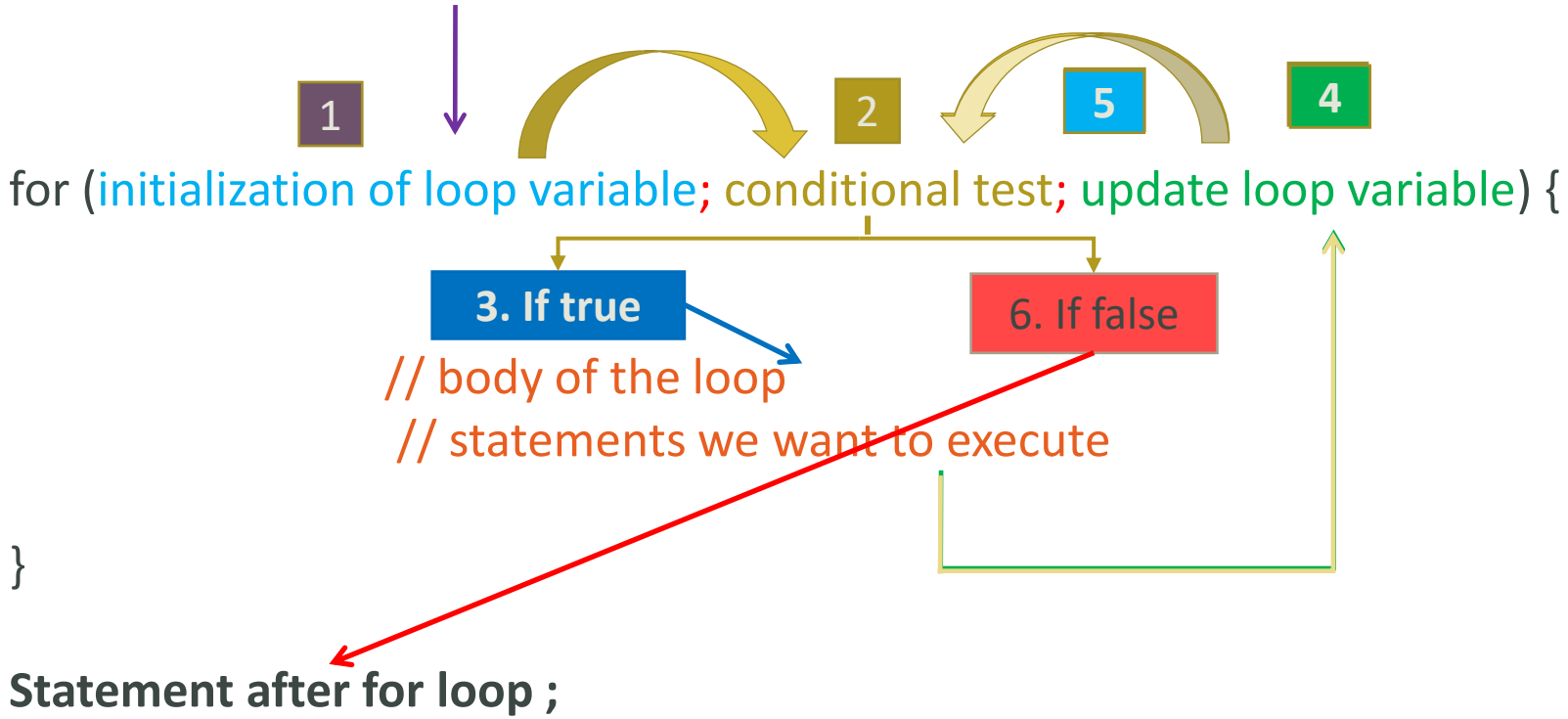
for Loop – Execution Steps

```
for (initialization of loop variable; conditional test; update loop variable) {

        // body of the loop

        // statements we want to execute

}
```

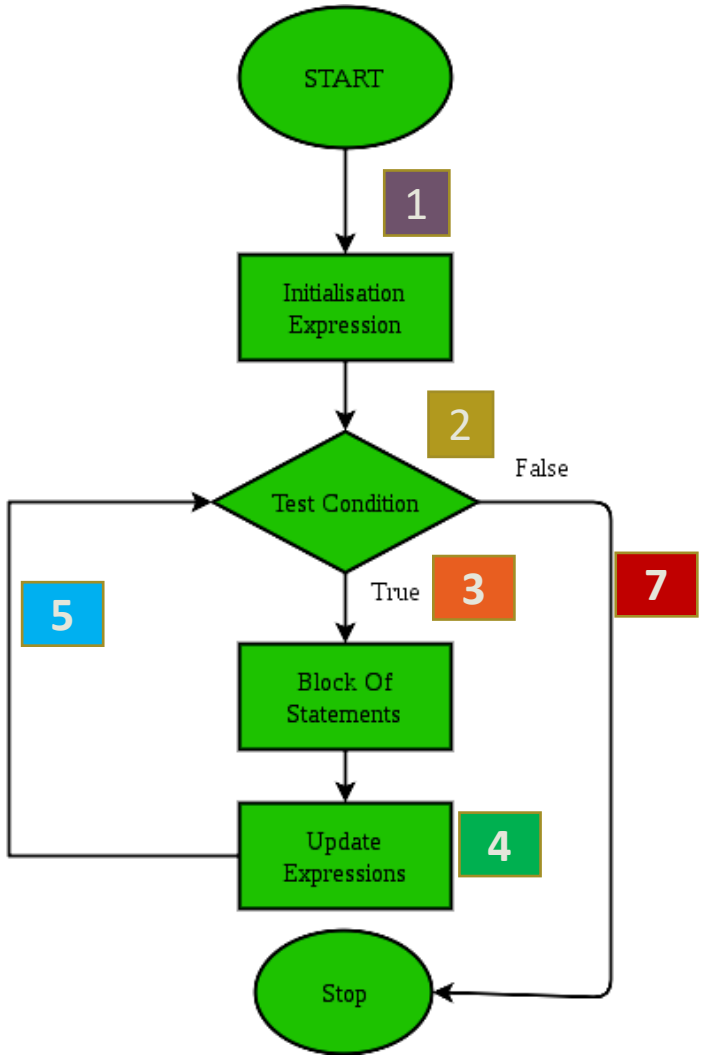
1. At first the loop variable is **initialized**. This statement is executed **only once.**
2. Then **condition** is **evaluated-**
    i. If the condition is **false**, it returns 0 and the **loop** is **terminated.**
    ii. if the condition is **true,** it returns 1, body of the loop is executed, loop variable is updated
3. Step 2 is **repeated until** the loop is **terminated**

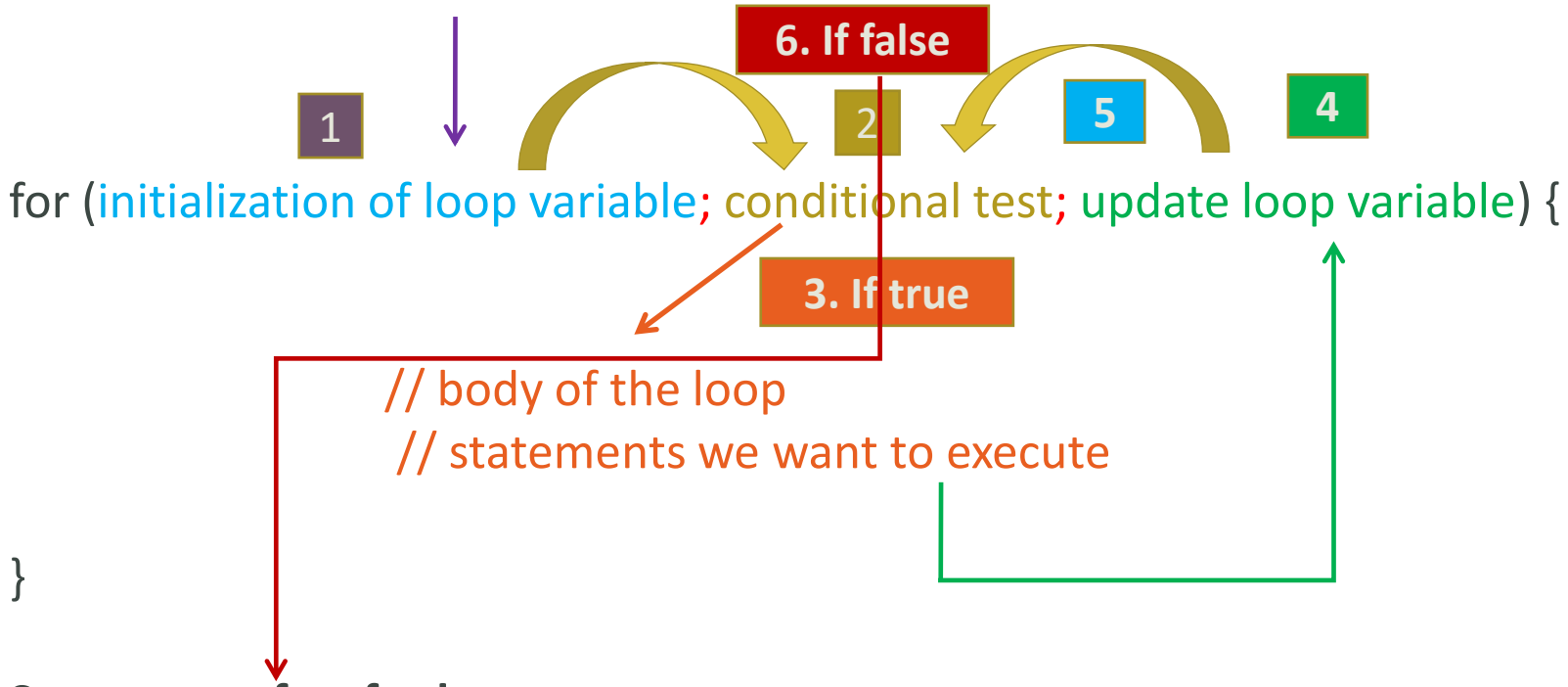
for Loop – Flow Chart





for Loop – Flow Chart

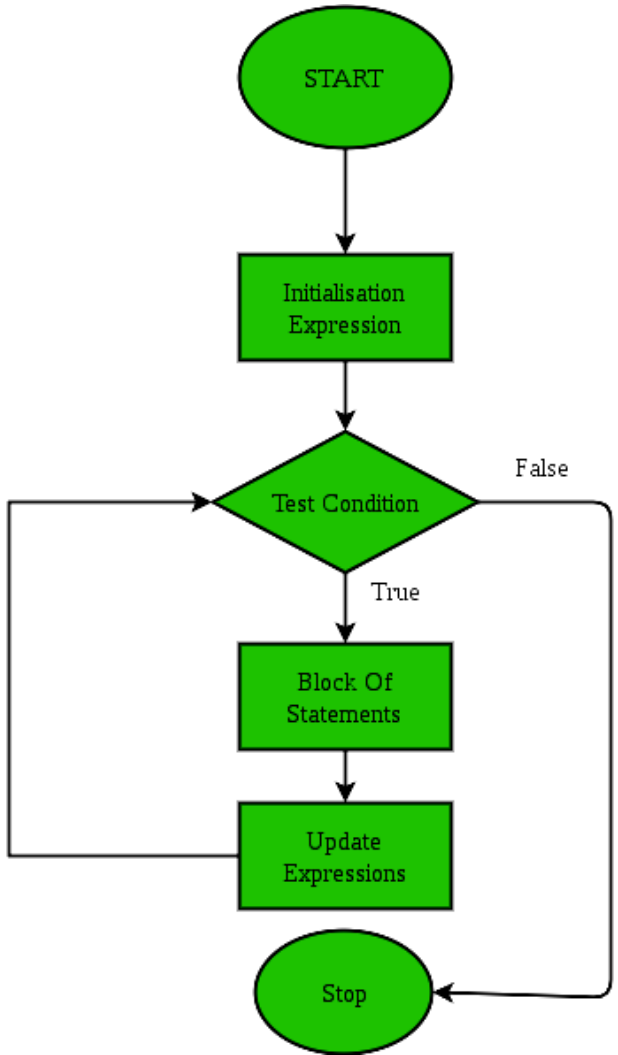




Statement after for loop ;

Terminate Loop and execute the statement right after the loop

Summary of for loop Execution



1. Initialize loop variable
2. Check condition
3. if false **terminate** loop body
4. If true **execute** loop body
5. Update loop variable and repeat from step 2

Sample for Loop

```
for (i=0 ; i<2 ; i++) {
        printf("Hello World");
}
printf("Outside loop body");
```

OUTPUT

Hello World
Hello World
Outside loop body

1. Here at first loop variable **i** is **initialized** with **0**
2. As step , condition **i<2** is **evaluated**. Since now value of **i is 0** which makes the condition **i<2 true**, the body of the for loop will be executed.
3. Then the loop variable i will be updated. As we have **i++,** value of i now will be 1
4. Condition **i<2** is **evaluated** and holds **true** for **i=1**, the body of the for loop will be executed
5. Then the loop variable i will be updated. As we have **i++,** value of i now will be 2
6. Condition **i<2** is **evaluated** and hold **false** for **i=2**, the body of the for loop will be terminated and statement immediately after loop body will be executed.

Sample for Loop

```
for (i=0 ; i<2 ; i++) {
        printf("Hello World");
}
printf("Outside loop body");
```

OUTPUT

Hello World
Hello World
Outside loop body

Value of i	Condition	State	Execution

Sample for Loop

- What should we change to print hello world 10 times?

```
for (i=0 ; i<10 ; i++) {
        printf("Hello World\n");
}
```

for Loop Example 1

- Print all the integer numbers from 1 to 10 inclusive
- What is the lowest number we need to print?
  - 1
- What is the largest number we need to print?
  - 10
- We can run the loop from 1, increase it by value 1 upto 10 and print each value

```
for(i=1; i<=10;i++){
        printf("%d\n",i);
}
```

for Loop Example 1

- **Print all the integer numbers from 1 to 10 inclusive**

```
for(i=1; i<=10;i++){
        printf("%d\n",i);
}
```

Value of i	Condition	State	Execution	Output
i=1	i<=10 or 1<=10	true	Loop body	1
i=2	i<=10 or 2<=10	true	Loop body	2
i=3	i<=10 or 3<=10	true	Loop body	3
i=4	i<=10 or 4<=10	true	Loop body	4
i=5	i<=10 or 5<=10	true	Loop body	5
i=6	i<=10 or 6<=10	true	Loop body	6
i=7	i<=10 or 7<=10	true	Loop body	7
i=8	i<=10 or 8<=10	true	Loop body	8
i=9	i<=10 or 9<=10	true	Loop body	9
i=10	i<=10 or 10<=10	true	Loop body	10
i=11	i<=10 or 11<=10	false	Outside loop	

for Loop Example 2

- **Print the sum of all the integer numbers from 1 to 5 inclusive**
- How many integer numbers are there between 1 to 5 inclusive?
  - 5
- How many numbers do we need to add?
  - 5 numbers
- What is the lowest number to add in the range?
  - 1
- What is the highest number to add in the range?
  - 5
- How many times should my loop iterate?
  - 5 times

```
int sum=0;
for(i=1; i<=5;i++){
        sum=sum+i;
}
printf("%d",sum);
```

for Loop Example 2

- **Print the sum of all the integer numbers from 1 to 5 inclusive**

```
int sum=0;
for(i=1; i<=5;i++){
        sum=sum+i;
}
printf("%d",sum);
```

Value of i	Condition	State	Execution	sum=sum+i
i=1	i<5 or 1<=5	true	Loop body	1
i=2	i<5 or 2<=5	true	Loop body	3
i=3	i<5 or 3<=5	true	Loop body	6
i=4	i<5 or 4<=5	true	Loop body	10
i=5	i<5 or 5<=5	true	Loop body	15
i=6	i<5 or 6<=5	false	Outside loop	

for Loop Example 3

- **Print the following series and its sum:**
- **1+2+3+4+5=15**

```
int sum=0;
for(i=1; i<=5;i++){
        printf("%d+",i);
       sum=sum+i;
}
printf("\b=%d",sum);
```

Value of i	Condition	State	Execution	sum= sum+i	Output
i=1	i<5 or 1<=5	true	Loop body	1	1+
i=2	i<5 or 2<=5	true	Loop body	3	2+
i=3	i<5 or 3<=5	true	Loop body	6	3+
i=4	i<5 or 4<=5	true	Loop body	10	4+
i=5	i<5 or 5<=5	true	Loop body	15	5+
i=6	i<5 or 6<=5	false	Outside loop		=15

for Loop Example 4

- **Print all the even numbers from 1 to 10 inclusive**
- How many even numbers are there between 1 to 10 inclusive?
  - 5
- How many times do you have to print?
  - 5 times
- How many times should my loop iterate?
  - 5 times
- What is the lowest number to print in the range?
  - 2
- What is the highest number to print in the range?
  - 10

```
for(i=1; i<=10;i++){
        if(i%2==0){
        printf("%d\n",i);
        }
}
```

for Loop Example 4

- **Print all the even numbers from 1 to 10 inclusive:**

```
for(i=1; i<=10;i++){
        if(i%2==0){
        printf("%d\n",i);
        }
}
```

Value of i	Condition	State	Execution	Output
i=1	i<10 or 1<=10	true	Loop body	
i=2	i<10 or 2<=10	true	Loop body	2
i=3	i<10 or 3<=10	true	Loop body	
i=4	i<10 or 4<=10	true	Loop body	4
i=5	i<10 or 5<=10	true	Loop body	
i=6	i<10 or 6<=10	true	Loop body	6
i=7	i<10 or 7<=10	true	Loop body	
i=8	i<10 or 8<=10	true	Loop body	8
i=9	i<10 or 9<=10	true	Loop body	
i=10	i<10 or 10<=10	true	Loop body	10
i=11	i<10 or 11<=10	false	Outside loop	

for Loop Example 5

- **Print the following series and its sum:**
- **2+4+6+8+10 = 30**

```
int sum=0;
for(i=1; i<=10;i++){
        if(i%2==0){
        printf("%d+",i);
       sum=sum+i;
       }
}
printf(“\b=%d”,sum);
```

Value of i	Condition	State	Execution	Sum=sum+i	Output
i=1	i<10 or 1<=10	true	Loop body		
i=2	i<10 or 2<=10	true	Loop body	2	2+
i=3	i<10 or 3<=10	true	Loop body		
i=4	i<10 or 4<=10	true	Loop body	6	4+
i=5	i<10 or 5<=10	true	Loop body		
i=6	i<10 or 6<=10	true	Loop body	12	6+
i=7	i<10 or 7<=10	true	Loop body		
i=8	i<10 or 8<=10	true	Loop body	20	8+
i=9	i<10 or 9<=10	true	Loop body		
i=10	i<10 or 10<=10	true	Loop body	30	10+
i=11	i<10 or 11<=10	false	Outside loop		

for Loop Example 6(Self Study)

- **Print all the odd numbers from 1 to 10 inclusive:**

```
for(i=1; i<=10;i++){
        if(i%2!=0){
        printf("%d\n",i);
        }
}
```

Value of i	Condition	State	Execution	Output
i=1	i<10 or 1<=10	true	Loop body	1
i=2	i<10 or 2<=10	true	Loop body	
i=3	i<10 or 3<=10	true	Loop body	3
i=4	i<10 or 4<=10	true	Loop body	
i=5	i<10 or 5<=10	true	Loop body	5
i=6	i<10 or 6<=10	true	Loop body	
i=7	i<10 or 7<=10	true	Loop body	7
i=8	i<10 or 8<=10	true	Loop body	
i=9	i<10 or 9<=10	true	Loop body	9
i=10	i<10 or 10<=10	true	Loop body	
i=11	i<10 or 11<=10	false	Outside loop	

for Loop Example 7(Self Study)

- **Print the following series and its sum:**
- **1+3+5+7+9 = 25**

```
int sum=0;
for(i=1; i<=10;i++){
        if(i%2!=0){
        printf("%d+",i);
       sum=sum+i;
       }
}
printf("\b=%d",sum);
```

Value of i	Condition	State	Execution	Sum=sum+i	Output
i=1	i<10 or 1<=10	true	Loop body	1	1+
i=2	i<10 or 2<=10	true	Loop body		
i=3	i<10 or 3<=10	true	Loop body	4	3+
i=4	i<10 or 4<=10	true	Loop body		
i=5	i<10 or 5<=10	true	Loop body	9	5+
i=6	i<10 or 6<=10	true	Loop body		
i=7	i<10 or 7<=10	true	Loop body	16	7+
i=8	i<10 or 8<=10	true	Loop body		
i=9	i<10 or 9<=10	true	Loop body	25	9+
i=10	i<10 or 10<=10	true	Loop body		
i=11	i<10 or 11<=10	false	Outside loop		

Infinite for loop

- As the name suggests, a loop that iterates for infinite time/ never stops iterating, is called infinite for loop
- **When an infinite loop occurs?**
  - Infinite loop occurs if the **condition** of the loop is always true
- **Example:**

```
for(i=1 ; i>0 ; i++){
        printf("print inside for loop");
}
```

Jump statements in for loop

- **Break Statement:** When a **break statement** is encountered inside a **loop/switch-case block**, the **loop/switch-case block** is immediately terminated and the program control resumes at the next statement after the **loop/switch-case block**

- **Continue Statement:** When **continue** is encountered inside any loop, the **remaining statement** of the loop body is **skipped** and **control** of the loop is passed to the next step – **update variable**

Jump statements in for loop Example

```
for(i=1; i<=3;i++){
        printf("%d\n",i);
        if(i==2){
                break;
        }
   printf("%d\n",i);
}
```

```
for(i=1; i<=3;i++){
        printf("%d\n",i);
        if(i==2){
                continue;
        }
   printf("%d\n",i);
}
```

Variations of for loop

- **Print all the integer numbers from 1 to 3**

We can initialize the loop variable before the loop statement.

```
int i=1;
for(; i<3;i++){
        printf("%d\n",i);
}
```

Value of i	Condition	State	Execution	Output
i=1	i<3 or 1<3	true	Loop body	1
i=2	i<3 or 2<3	true	Loop body	2
i=3	i<3 or 3<3	false	Outside loop	

Variations of for loop

- **Print all the integer numbers from 1 to 3**

We can update the loop variable inside the loop body.

```
int i=1;
for(; i<3;){
        printf("%d\n",i);
        i++;
}
```

Value of i	Condition	State	Execution	Output
i=1	i<3 or 1<3	true	Loop body	1
i=2	i<3 or 2<3	true	Loop body	2
i=3	i<3 or 3<3	false	Outside loop	

Task – Solve Using for loop

- **Print the following series up to n and its sum** :
- 1 + 2 + 3 + … + n
- 0 + 2 + 4 + … + n
- 1+ 3 + 5 + … + n
- 1 - 2 + 3 - 4 + … n
- $1^2 + 2^2 + 3^2 + … + n^2$
- x + 2x^2 + 3x^3 + ….. + nx^n

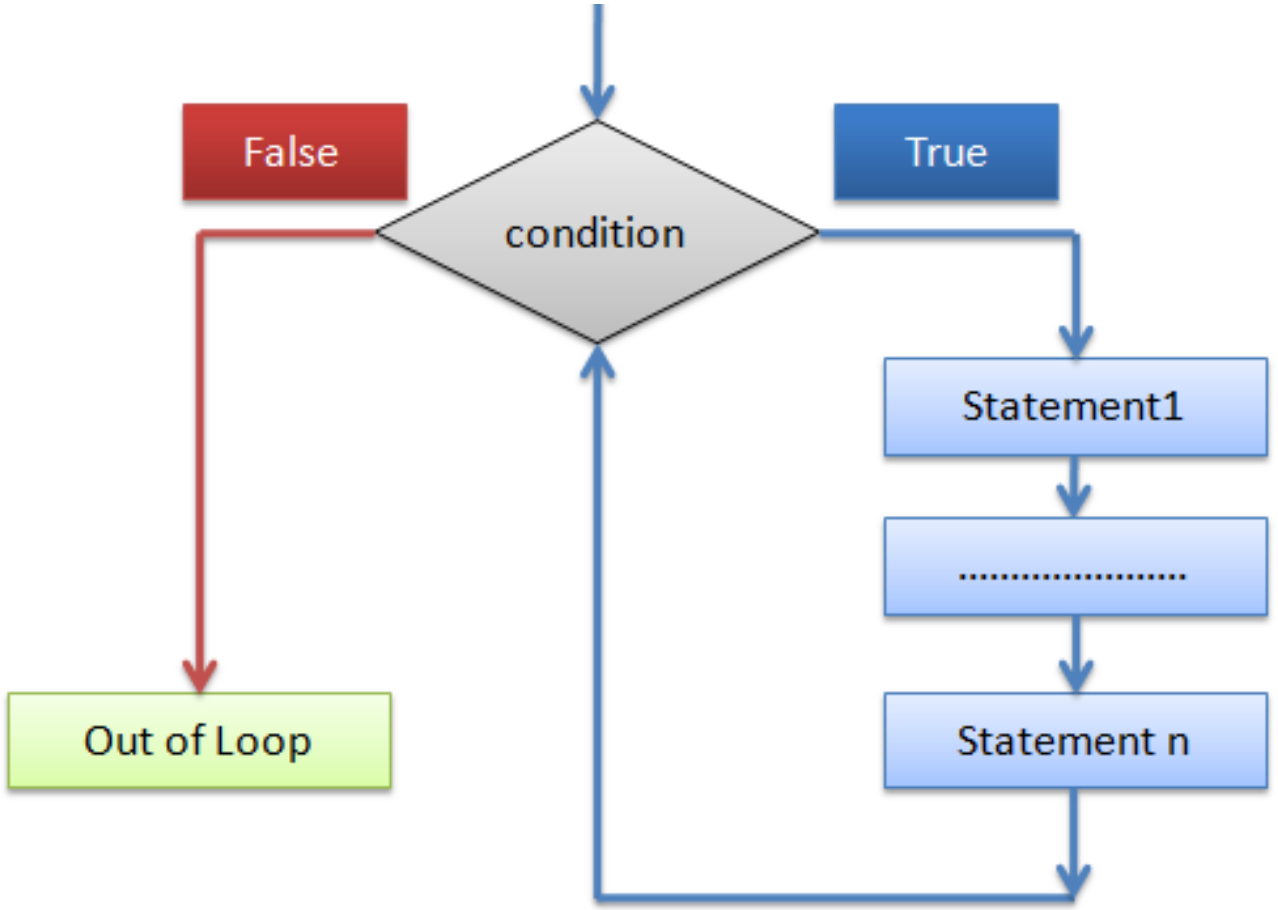
Task - Solve Using for loop

- Print the alphabets from A to Z
- Print the alphabets from a to z
- Print the alphabets from a to z without the vowels
- Take input of a number **n** and print all of its factors
- Take input of two numbers **a** & **b,** and print their GCD(Greatest common divisor)
- Take input of a number and print its factorial

while Loop – Syntax & Flow Chart

- **Syntax:**

```
while (conditional test) {
        statement 1;
        statement 2;
        ……
        statement n;
}
```

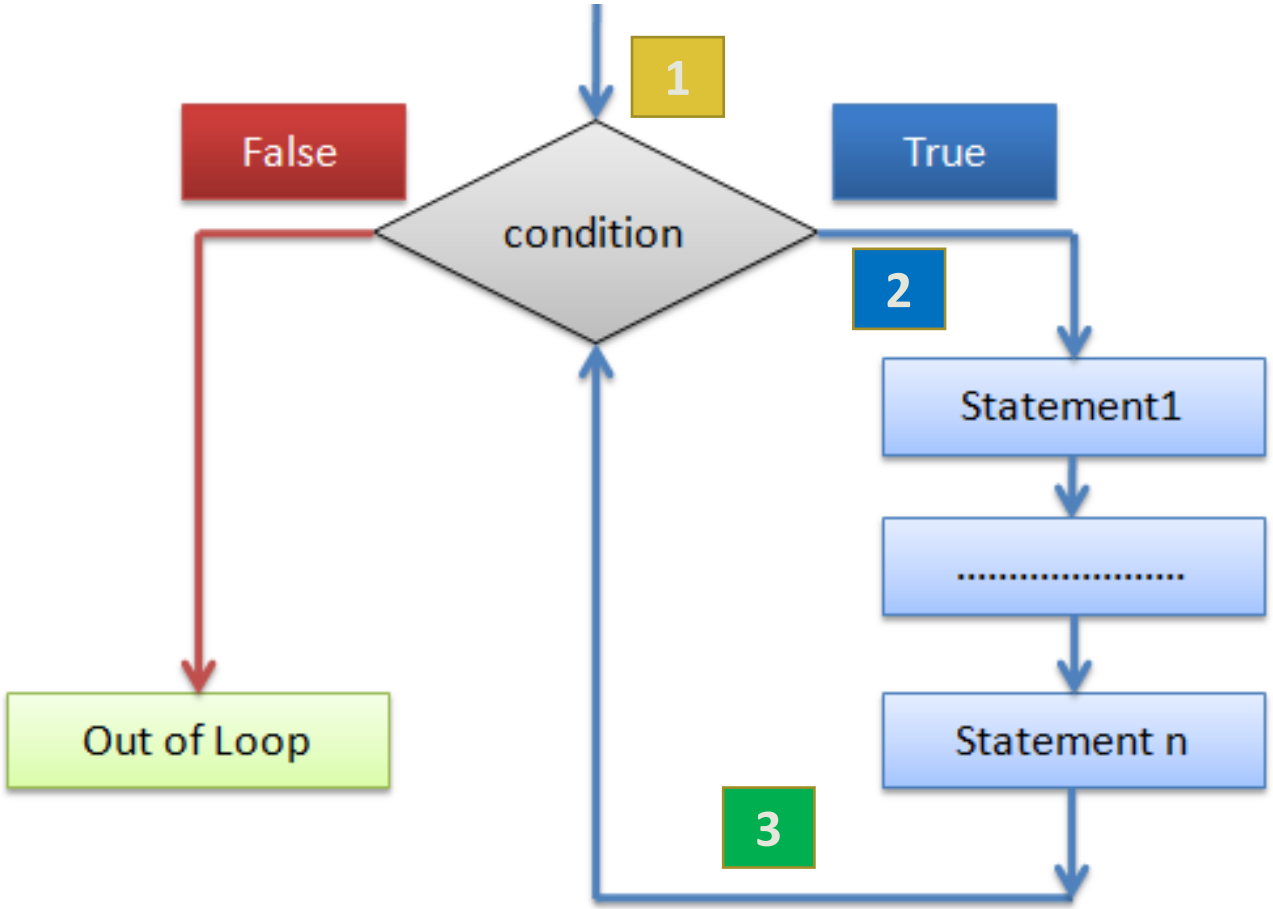


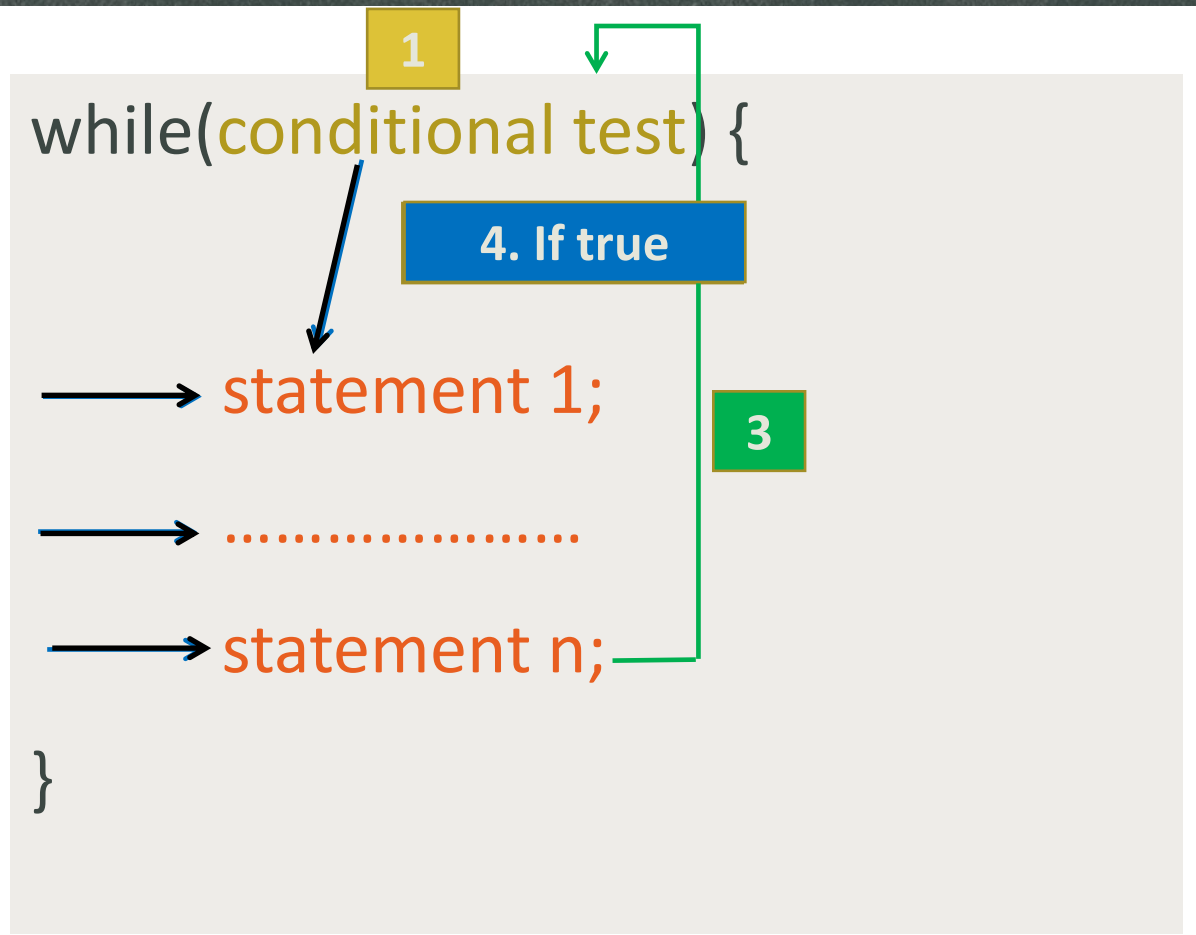
while Loop – Execution Steps

```
while(conditional test) {

        // body of the loop

         // statements we want to execute

}
```

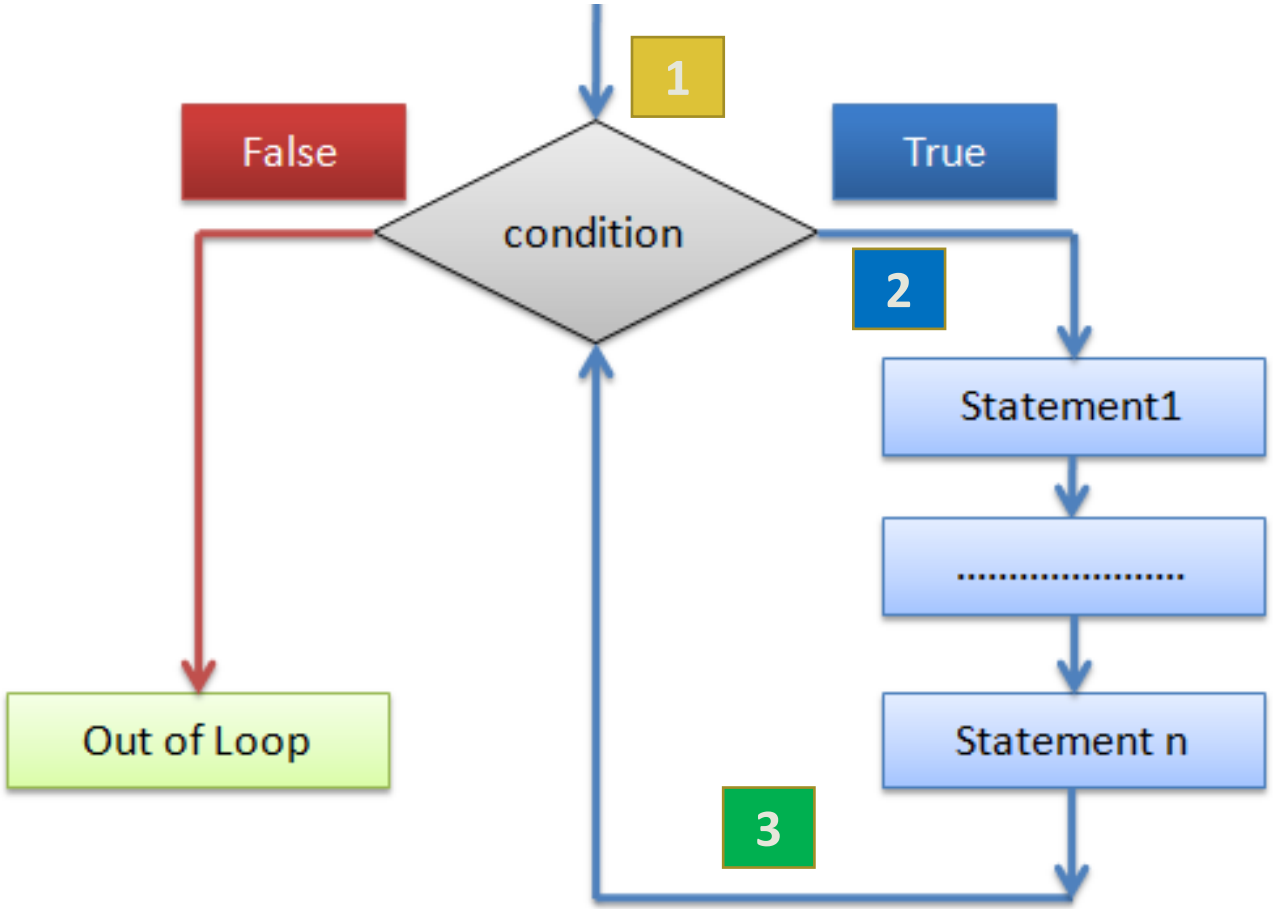
1. **Condition** is **evaluated-**
    i. If the condition is **false**, it returns 0 and the **loop** is **terminated.**
    ii. if the condition is **true,** it returns 1, body of the loop is executed
2. **Step 1 is repeated until the condition is false and loop is terminated**

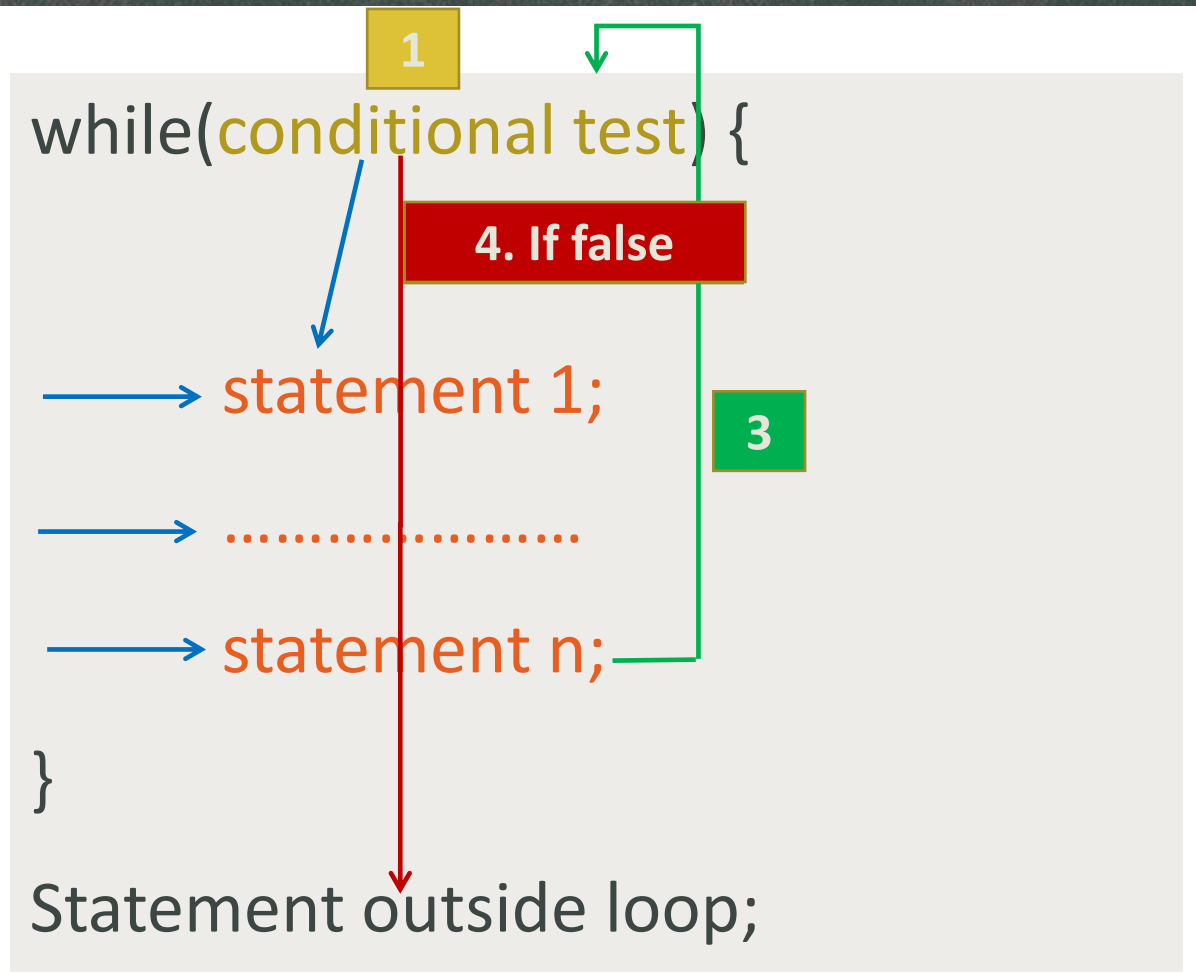
while Loop – Execution Simulation



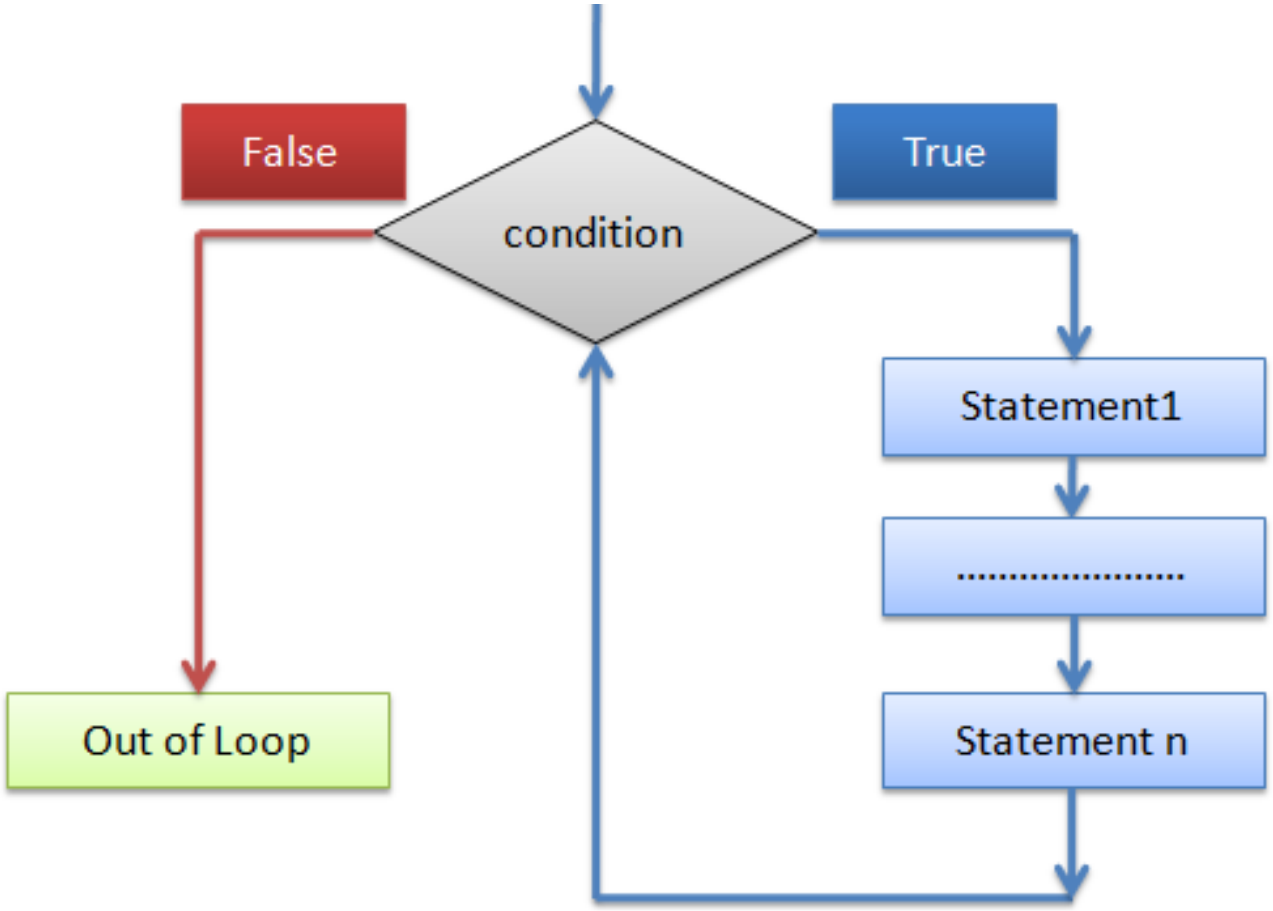


while Loop – Execution Simulation





Summary of while loop Execution



1. Check condition
2. if false **terminate** loop body
3. If true **execute** loop body
4. repeat from step 1

Sample while Loop

```
1.  int i = 0;
2.  while  ( i<2) {
3.      printf("Hello World\n");
4.      i++;
5.  }
6.  printf("Outside loop body");
```

OUTPUT

Hello World
Hello World
Outside loop body

1. Here at line 1 **i** is **initialized** with **0**
2. Condition **i<2** is **evaluated**. Since now value of **i is 0** which makes the condition **i<2 true**, the body of the while loop will be executed.
3. At line 4, as we have **i++,** value of i now will be 1
4. Condition **i<2** is **evaluated** and hold **true** for **i=1**, the body of the while loop will be executed
5. Again at line 4, we have **i++,** value of i now will be 2
6. Condition **i<2** is **evaluated** and hold **false** for **i=2**, the body of the while loop will be terminated and statement immediate after loop body will be executed.

Sample while Loop

```
1. int i = 0;
2. while  ( i<2) {
3.       printf("Hello World");
4.       i++;
5. }
6. printf("Outside loop body");
```

Value of i	Condition	State	Execution

OUTPUT

Hello World
Hello World
Outside loop body

while Loop Example 1

- **Print all the integer numbers from 1 to 10 inclusive**

```
for(i=1; i<=10;i++){
        printf("%d\n",i);
}
```

```
int i=1;
while(i<=10){
        printf("%d\n",i);
        i++;
}
```

while Loop Example 2

- **Print all the odd numbers from 1 to 10 inclusive**

```
for(i=1; i<=10;i++){
        if(i%2!=0){
        printf("%d\n",i);
        }
}
```

```
int i=1;
while(i<=10){
        if(i%2!=0){
        printf("%d\n",i);
        }
        i++;
}
```

Task – Solve Using while loop

Print the following series up to n and its sum :

1 + 2 + 3 + … + n

0 + 2 + 4 + … + n

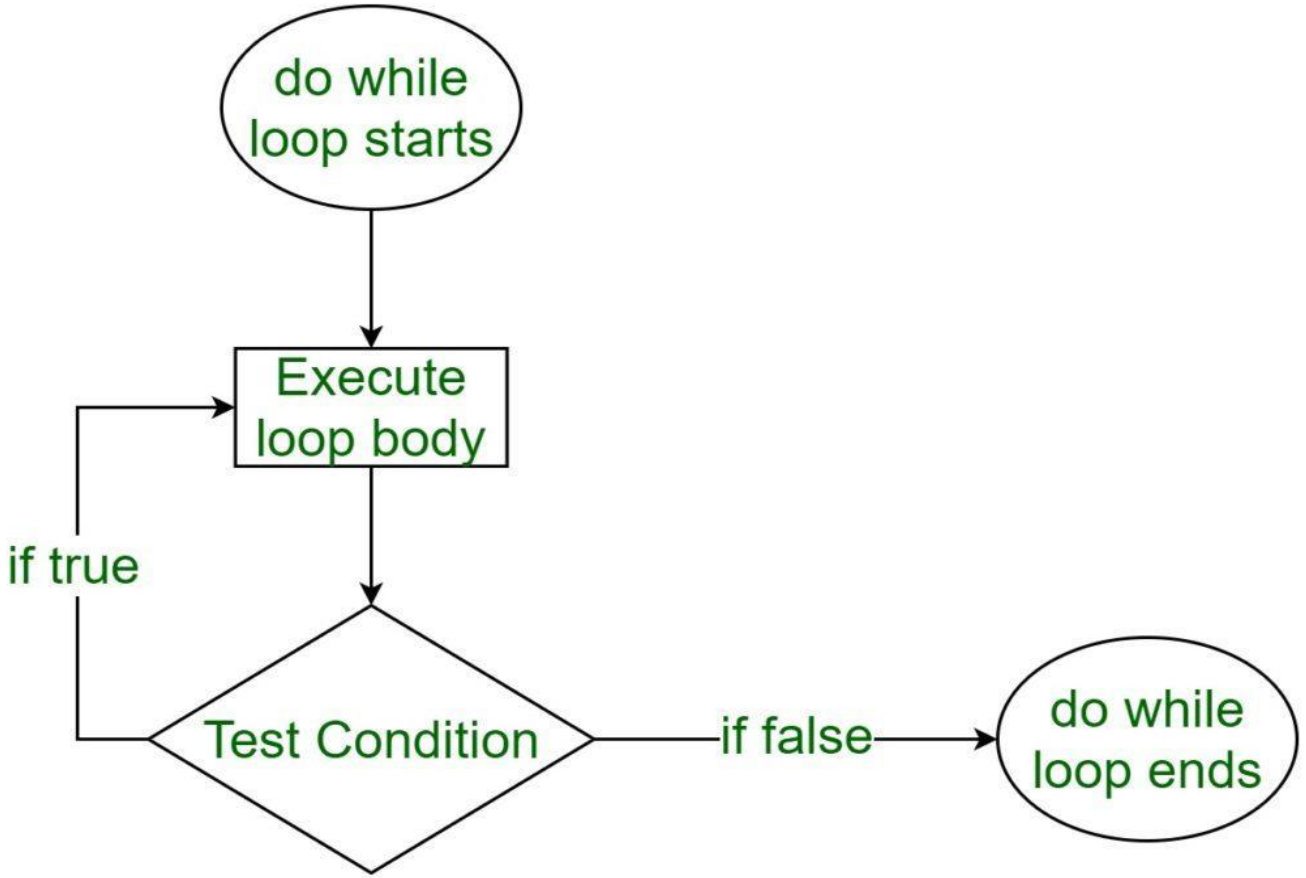
1+ 3 + 5 + … + n

- Print all the even numbers from 1 to 10 inclusive
- Print the reverse of a given number
- Take input of a number **n** and print all of its factors

do while Loop – Syntax & Flow Chart

- **Syntax:**

```
do{
        statement 1;
        statement 2;
        ……
        statement n;
} while (conditional test) ;
```

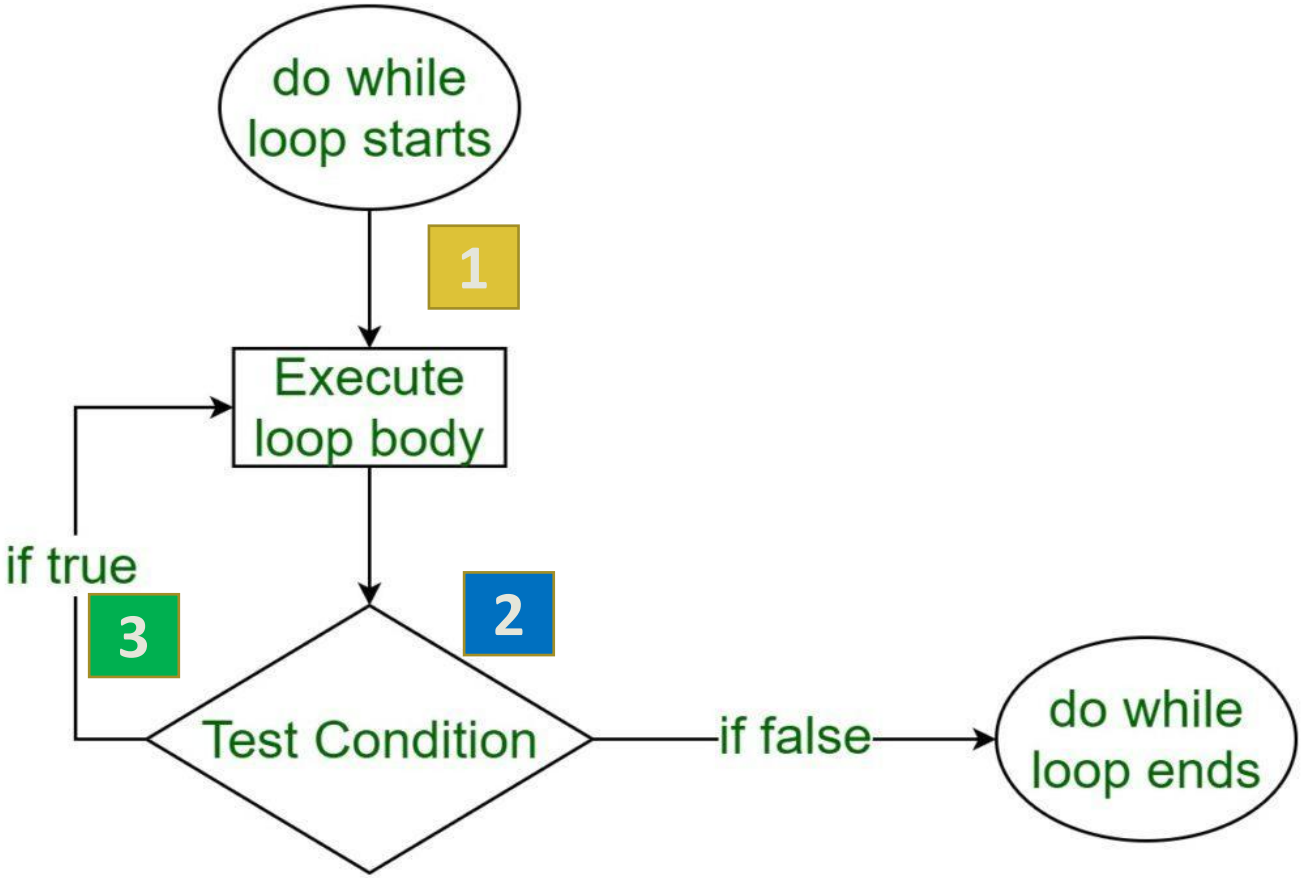


do while Loop – Execution Steps

```
do{
        // body of the loop
        // statements we want to execute
} while(conditional test) ;
```

1. **Body of the loop is executed first.**
2. **Condition** is **evaluated-**
    i. If the condition is **false**, it returns 0 and the **loop** is **terminated.**
    ii. if the condition is **true,** it returns 1, body of the loop is executed
2. **Step 2 is repeated until the condition is false and loop is terminated**

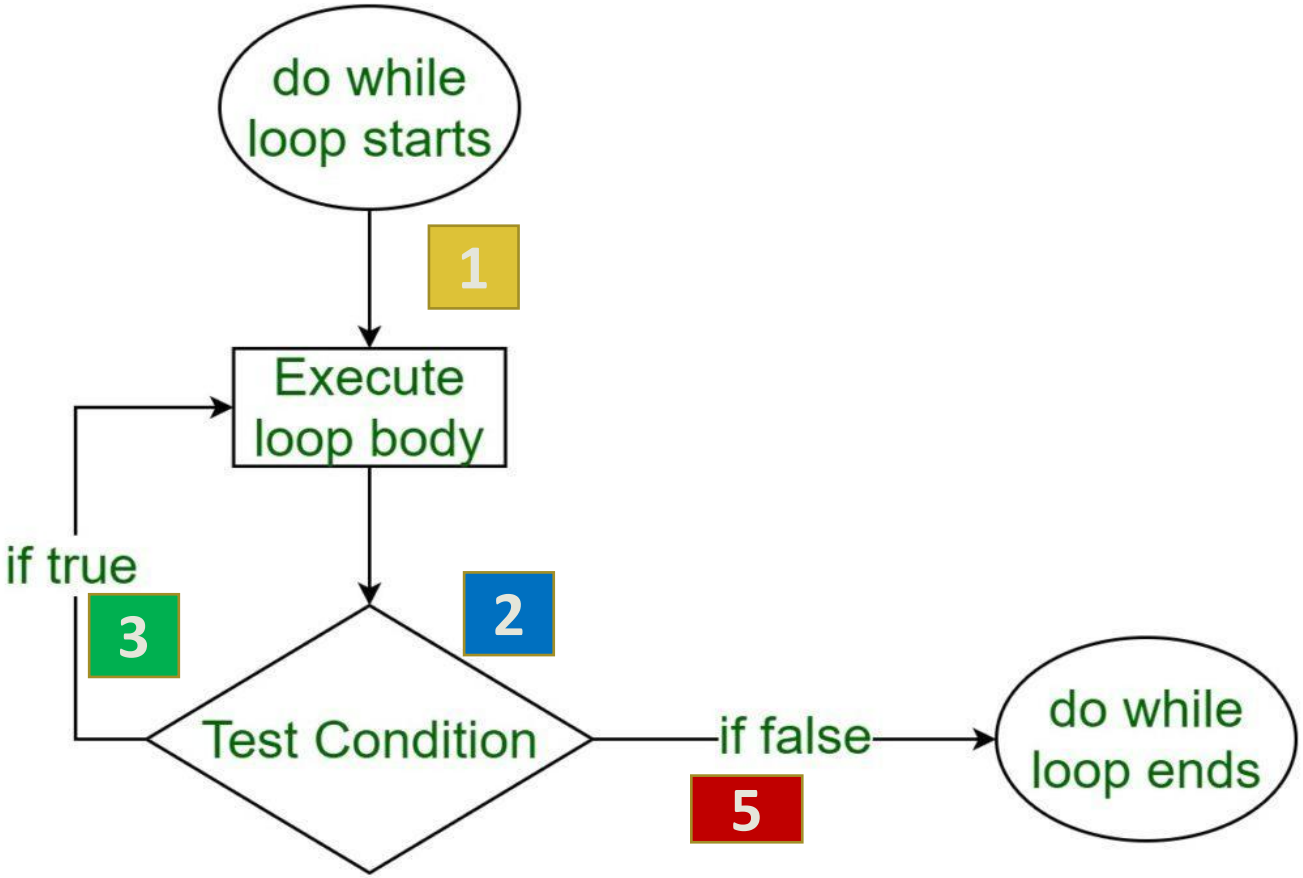
do while Loop – Execution Simulation




```
do{   1

   ⟶statement 1;
   ⟶.....
   ⟶statement n;
} while(conditional test) ;  4
```

5. If true

do while Loop – Execution Simulation




```
do{ 1

    statement 1;
    ..................
    statement n;

} while(conditional test) ; 4
Statement outside loop;
```

3. If true

5. If false

Sample do while Loop

```
1. int i = 0;
2. do{
3.       printf("Hello World");
4.       i++;
5. } while  ( i<2) ;
6. printf("Outside loop body");
```

Execution	Value of i	Condition	State
Loop body	i=1	i<2 or 0<2	true
Loop body	i=2	i<2 or 1<2	false
Outside loop			

OUTPUT

```
Hello World
Hello World
Outside loop body
```

Difference Between while and do while Loop

- In while loop, find the condition is checked, then the body is executed if condition is true.
- In do...while loop, first the body is executed, then condition is checked.
- the body of do...while loop will execute at least once irrespective to the test-expression.

Difference Between while and do while Loop

```
int i=0;
while(i<0){
  printf("I is a positive number");
  i++;
}
```

```
int i=0;
do{
printf("I is a negative number");
  i++;
}while(i<0);
```

Difference Between while and do while Loop

```
int i=0;
while(i<0){
   printf("I is a positive number");
   i++;
}
```

Value of i	Condition	State	Execution
i=0	i<0 or 0<0	false	Outside loop

Difference Between while and do while Loop

```
int i=0;
do{
printf("I is a negative number");
   i++;
}while(i<0);
```

Output:

I is a negative number

Execution	Value of i	Condition	State
Loop body	i=1	i<0 or 1<0	false
Outside loop			


Thank
you!